

Clustering Analysis of La Liga Football Teams (2015/16)

Data Mining – Trento University

Cristina Morillo Leal

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ABSTRACT

This study applies clustering techniques to the performance data of La Liga teams during the 2015/16 season, with the aim of identifying meaningful patterns and significant groupings. Using K-Means, Gaussian Mixture Model, and agglomerative hierarchical clustering, teams are classified based on a range of statistical indicators related to attacking performance, build-up play, and defensive behavior. The analysis reveals that variables such as shots, completed passes, and duels won are the most influential features driving cluster formation, playing a key role in distinguishing between different team profiles. Results indicate that build-up play metrics predominantly shape the clusters, providing valuable insights into team performance trends and highlighting the usefulness of clustering in football analytics.

Keywords: Football, Cluster, KMeans, Gaussian Mixture, Agglomerative Hierarchical

1. INTRODUCTION

Advanced data analysis has become a key tool for understanding sports performance beyond final results. In this project, clustering techniques are applied to analyze La Liga teams during the 2015/2016 season, with the aim of identifying groups of teams with similar playing characteristics and performance profiles. Using algorithms such as K-Means, Gaussian Mixture Models, and Agglomerative hierarchical clustering, the study seeks to uncover underlying patterns in the data that allow for an alternative segmentation of teams beyond the traditional league table, providing deeper insight into the competitive dynamics of the league.

For a step-by-step explanation of the analysis and its outcomes, the interactive notebooks are available at the links below:

- **Colab Notebook 1: Data Collection and EDA** - https://colab.research.google.com/drive/1oPzQyoIxA_f6vECo_9JGAFpBsP4UaaCe?usp=sharing
- **Colab Notebook 2: Clustering analysis** - https://colab.research.google.com/drive/1NN5Wfrk4W59n2UPYHpDPRVuXFt_PFDtk?usp=sharing

2. METHODOLOGY

2.1 Data source: StatsBomb Open Data

This study uses event-level football data from StatsBomb, a leading sports analytics provider renowned for the depth and quality of its datasets. The analysis is based on StatsBomb’s open-access data, which offers detailed match-level information well suited for advanced data science and analytical research [1].

The data used in this project corresponds to the Spanish top division (LaLiga) for the 2015/2016 season, covering all matches played during the competition. Two types of data were collected:

- **Event-level data**, containing detailed information about on-ball and off-ball actions occurring during matches, such as passes, shots, duels, pressures, recoveries, and fouls. Each event is associated with spatial coordinates, timestamps, players, and contextual attributes.
- **Match-level metadata**, providing contextual information for each match, including participating teams, match identifiers, and competition details.

Data were retrieved using the official StatsBomb Python library and merged at the event and match level using a unique match identifier.

2.2 Variable definition

Based on the unified match and event dataset, team-level features were defined to describe playing styles across different phases of the game. The variables were aggregated at team level and organised into three datasets corresponding to the attacking, build-up, and defensive phases of play.

Attacking phase variables

Attacking performance was described using a combination of volume-based and efficiency-based metrics. Absolute attacking output was captured through per-90-minute variables such as shots, shots on target, goals, expected goals (xG), and their corresponding values for shots taken outside the box. In parallel, several relative metrics were computed to reflect attacking efficiency, including goals per shot, goals per shot on target, xG per shot, and the proportion of shots, shots on target, and goals generated from outside the penalty area.

Build-up phase variables

The build-up phase focuses on passing behaviour and ball circulation. Absolute metrics per 90 minutes were used to quantify passing volume, including total passes, completed passes, final-third passes, short and long passes, and crosses. In addition, several relative variables were computed to capture stylistic and qualitative aspects of build-up play, such as the proportion of short, long, final-third passes and crosses, as well as overall and type-specific pass completion percentages. Average pass length was also included to provide an overall indication of directness in possession.

Defensive phase variables

Defensive behaviour was described through both activity and effectiveness metrics. Absolute per-90-minute variables include recoveries, recoveries in the final third, duels, duels won, fouls committed, shots

conceded, shots on target conceded, expected goals against (xGA), and goals conceded. Complementary percentage-based variables were computed to capture defensive efficiency and risk, such as duel success rate, proportion of recoveries and fouls occurring in the final third, and the relative share of shots, shots on target, xGA, and goals conceded. Clean sheets were also included as a high-level defensive outcome indicator.

Across all phases, absolute metrics were normalised per 90 minutes to ensure comparability between teams and to facilitate intuitive visual interpretation during exploratory analysis. Relative metrics expressed as percentages or ratios were additionally computed, as they better capture efficiency and stylistic tendencies and are more suitable for distance-based clustering methods.

2.3 Clustering algorithms

In this section, different clustering methods were explored to uncover underlying patterns and segment the data into meaningful groups. Three different algorithms—K-Means, Gaussian Mixture Models, and Hierarchical Clustering—were evaluated.

2.3.1 K-Means Clustering

K-Means clustering is an unsupervised learning algorithm whose objective is to partition a dataset into K distinct, non-overlapping clusters such that observations within each cluster are as similar as possible, typically measured by Euclidean distance. Formally, K-Means minimizes the within-cluster sum of squared distances between data points and their assigned cluster centroids.

From a general perspective, the algorithm proceeds iteratively. First, K centroids are initialized, often at random. Each data point is then assigned to the nearest centroid, forming K clusters. Next, the centroids are updated as the mean of all points assigned to each cluster. These assignment and update steps repeat until convergence, which occurs when cluster assignments no longer change or when the decrease in the objective function becomes negligible.

K-Means offers several advantages: it is conceptually simple, computationally efficient, and scales well to large datasets. However, it also has notable limitations. The algorithm requires the number of clusters K to be specified in advance, is sensitive to initialization and outliers, and performs poorly when clusters are non-spherical or have unequal sizes and densities [2].

Because the performance of K-Means is highly dependent on the choice of K , selecting an appropriate number of clusters becomes a critical step in the clustering process. The Elbow Method is a commonly used heuristic for this purpose. It consists of running K-Means for different values of K and plotting the within-cluster sum of squares against K . As K increases, this metric decreases, but with diminishing returns. The “elbow” point—where the rate of decrease sharply changes—is interpreted as a reasonable trade-off between model complexity and clustering quality [3].

2.3.2 Gaussian Mixture Models

Gaussian Mixture Models (GMMs) are a probabilistic approach to clustering that model data as a weighted combination of multiple Gaussian distributions. The main objective of a GMM is to represent

the underlying data distribution and identify latent subpopulations (clusters) by estimating the parameters (means, covariances, and mixing coefficients) of each Gaussian component. Unlike hard clustering methods, GMMs perform probabilistic clustering, where each data point is assigned a probability of belonging to each cluster rather than a single deterministic label.

Broadly speaking, GMMs are typically trained using the Expectation–Maximization (EM) algorithm. In the Expectation (E) step, the model computes the posterior probability that each data point belongs to each Gaussian component, given the current parameter estimates. In the Maximization (M) step, the parameters of the Gaussians are updated to maximize the expected log-likelihood of the data under these probabilistic assignments. These two steps are iterated until convergence.

A key advantage of GMMs is their flexibility: by allowing full covariance matrices, they can model clusters with different sizes, orientations, and elliptical shapes. Additionally, soft assignments are well suited to overlapping clusters and uncertainty modeling. However, GMMs can be computationally expensive, especially in high dimensions, and are sensitive to initialization, often converging to local optima. Their performance also depends on the validity of the Gaussian assumption, which may not hold for all datasets [2].

To select the optimal number of components, model selection criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are commonly used. Both criteria balance goodness of fit (log-likelihood) against model complexity, penalizing models with more parameters. AIC generally favors more complex models, while BIC imposes a stronger penalty, often leading to more parsimonious solutions. The model with the lowest AIC or BIC is typically selected as the best trade-off between fit and complexity [4].

2.3.3 Hierarchical Clustering

Hierarchical clustering is an unsupervised learning technique whose main objective is to identify a nested structure of groups within a dataset without requiring a predefined number of clusters. It produces a hierarchy of clusters that can be visualized using a dendrogram, enabling analysis at different levels of granularity.

From a high-level perspective, hierarchical clustering operates by successively grouping or splitting observations based on a chosen measure of similarity or distance. In agglomerative hierarchical clustering, which is the approach used in this project, each observation starts as its own cluster, and the algorithm iteratively merges the two most similar clusters until all observations belong to a single cluster. In contrast, divisive hierarchical clustering follows a top-down strategy, starting with all observations in one cluster and recursively splitting them, although this approach is less commonly used due to its higher computational cost.

One key advantage of hierarchical clustering is its interpretability: the dendrogram provides a transparent representation of the clustering process and does not require specifying the number of clusters in advance. However, the method suffers from high computational complexity, typically $O(n^2)$ in both time and memory, which limits its scalability to large datasets. Additionally, hierarchical clustering is sensitive to noise and outliers, as early erroneous merges cannot be corrected [2].

In this work, agglomerative hierarchical clustering was implemented using the Ward linkage method. At each step, this approach merges the pair of clusters that results in the minimum increase in total within-cluster variance, thereby favoring compact and homogeneous clusters. This variance-minimization principle often leads to well-balanced and interpretable clustering solutions [5].

2.4 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical technique for dimensionality reduction that aims to simplify complex datasets by transforming a set of original, potentially correlated variables into a smaller set of uncorrelated variables known as principal components. Each principal component is a linear combination of the original variables and is constructed to capture the maximum possible variance in the data, with components ordered according to the proportion of variance they explain.

By projecting the data onto a lower-dimensional space while preserving most of its variability, PCA reduces redundancy and noise, making the underlying structure of the data easier to analyze and interpret. This property is particularly useful when working with high-dimensional datasets, where direct visualization and interpretation are challenging [6].

In this study, PCA is employed exclusively for visualization purposes, reducing the original feature space to two principal components. This two-dimensional representation enables an effective graphical visualization of the clustering results, facilitating the assessment of cluster separation, compactness, and overall structure.

3. EXPLORATORY DATA ANALYSIS (EDA)

The following section analyses team-level metrics across attacking, build-up, and defensive phases to identify relevant performance patterns.

Attacking phase

The analysis of attacking based on the distribution of the variables (Figure 1) shows that the number of shots is relatively high and displays noticeable variability across teams, reflecting different attacking intensities. Shots on target are more concentrated, suggesting similar levels of shooting accuracy. Goals scored and xG are tightly grouped at low values, indicating limited differences in overall scoring output. Actions from outside the box show lower and more concentrated distributions, both in terms of shots and shots on target, highlighting their secondary role in most attacking strategies. Finally, goals scored from outside the box are rare and highly concentrated near zero, underlining that most goals come from inside the penalty area and that long-range goals are more exceptional than systematic.

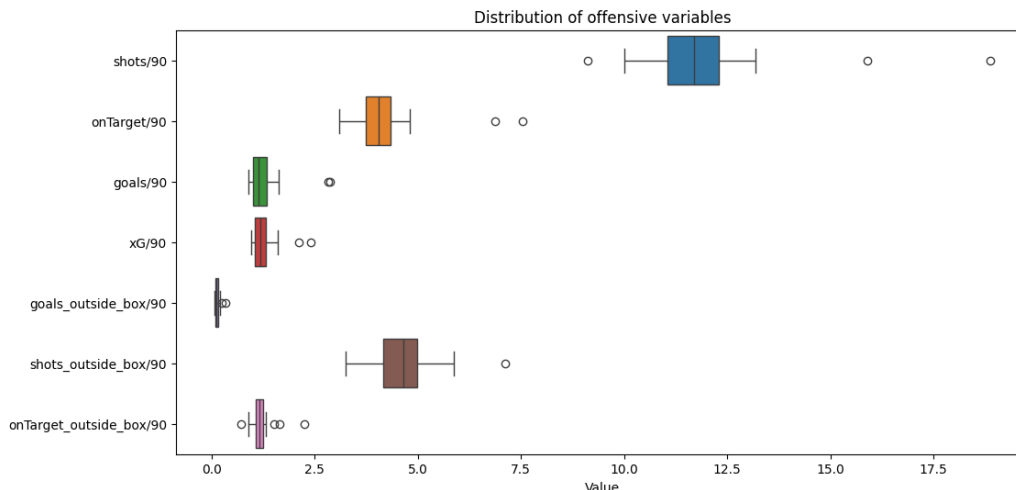


Figure 1. Distributions of Attacking Variables

At the team level (Figure A.1), distinct attacking profiles emerge. Real Madrid and Barcelona clearly dominate, combining high shot volumes with strong accuracy and elevated xG. Sevilla, Atlético de Madrid, and Rayo Vallecano also show consistent attacking involvement, albeit with slightly lower efficiency. Some teams, such as RC Deportivo de La Coruña and Granada, rely more on long-range shooting, whereas Barcelona and Atlético de Madrid focus their attacking play inside the box. Conversely, Granada, Getafe, and Sporting de Gijón exhibit more limited and reactive offensive patterns, with low values across most attacking metrics.

Scatter-based analyses (Figure A.2) further highlight that similar goal outputs can be achieved through markedly different strategies. While some teams depend on high shooting frequency, others convert fewer chances more efficiently, underscoring structural differences in attacking efficiency that are not captured by goal totals alone.

Build-up phase

The distribution of the variables in the build-up analysis (Figure 2) suggests that the total number of passes and completed passes are high and display noticeable variability across teams, reflecting different possession-based styles. The average pass length is highly concentrated, indicating similar passing distances among most teams. Short passes and short passes completed present moderate dispersion, suggesting differences in tempo and ball circulation. Passing activity in the final third, both attempted and completed, is lower and more tightly grouped, highlighting the difficulty of progressing the ball into advanced areas. Long passes show limited variability, while their completion rates are slightly more dispersed, reflecting differing risk profiles. Finally, the number of crosses and crosses completed are low and highly concentrated, indicating that crossing is a more situational and less dominant build-up mechanism across teams.

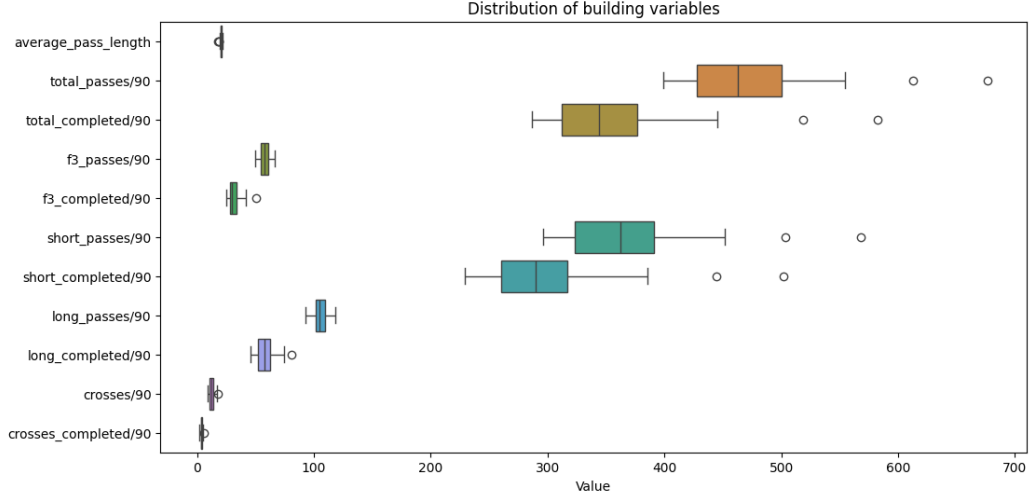


Figure 2. Distributions of Build-Up Variables

Clear differences in playing styles can be observed when examining passing metrics at team level (Figure A.3). Barcelona and Real Madrid stand out for a control-based model built on high passing volume and short-pass accuracy. Atlético de Madrid, Athletic Club, and Real Sociedad adopt a more vertical and efficient approach, with lower volume but greater progression into the final third. Wide play is particularly relevant for teams such as Sevilla and Rayo Vallecano, characterized by a high number of crosses. Meanwhile, RC Deportivo de La Coruña and Eibar prioritize long passes as their main method of advancement. Finally, teams such as Granada, Getafe, and Sporting de Gijón display a more reactive style, with low passing volume and a reliance on quick transitions.

The scatter plots (Figure A.4) reinforce the patterns previously identified in the distributional analysis. Teams with possession-oriented models concentrate in areas associated with high passing volume and short-pass usage, while more vertical or reactive sides are positioned in regions reflecting lower volume and greater directness.

Defensive phase

An analysis of defensive performance, based on the distribution of the variables (Figure 3), reveals that ball recoveries and defensive duels show consistently high values across teams, reflecting an overall intense defensive involvement. However, while the volume of duels varies more between teams, duels won exhibit lower dispersion, indicating that duel efficiency is more stable than duel frequency. Fouls committed minutes display greater variability and some outliers, reflecting different defensive styles, while fouls committed in the final third are low and tightly concentrated. Meanwhile, the number of shots conceded and shots on target conceded per match show clear differences between teams, helping to distinguish more solid defenses from weaker ones. Finally, expected goals against and goals conceded per match are concentrated at low values, and the number of clean sheets stands out for its high dispersion, making it a key variable for identifying top-level defenses.

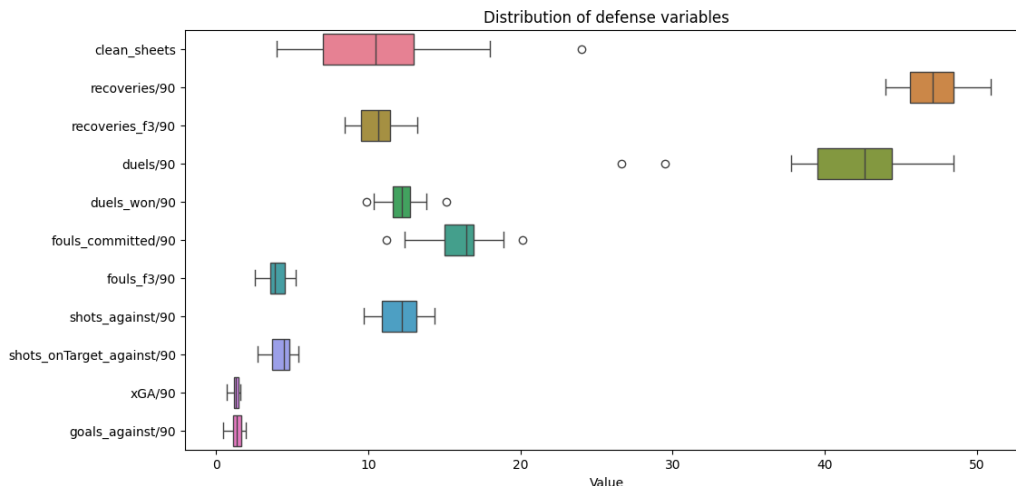


Figure 3. Distributions of Defensive Variables

Defensive metrics at team level (Figure A.5) underline marked variations in how teams approach defending. Atlético de Madrid stands out as the defensive benchmark, combining high ball recoveries in advanced areas with low shots against, xGA, and goals conceded, resulting in the highest number of clean sheets. Barcelona and Real Madrid also show strong defensive performance. Teams such as Eibar, Athletic Club, and Celta de Vigo display more aggressive defensive profiles, ranking highly in duels and recoveries, reflecting proactive and physically intense defending, though not always with low xGA. In contrast, Sporting de Gijón, Getafe, and Granada concede higher volumes of shots and xGA, suggesting more reactive, deep-block defensive styles. Foul metrics further differentiate approaches: teams with higher foul counts compensate for positional disadvantages through physical disruption, while low-foul teams rely more on structure and anticipation.

The scatter plots (Figure A.6) further strengthen these findings by revealing clear relationships between defensive volume, efficiency, and outcomes. Teams combining high recoveries and duel success with lower shots and xG conceded tend to lie below the diagonal, highlighting more effective defensive performance rather than mere activity. In contrast, sides with aggressive defensive involvement but higher shots or xG against illustrate that intensity alone does not guarantee solidity.

Cross phase

The combined analysis (Figure A.7) of attacking, build-up and defensive metrics highlights clear stylistic differences across teams. Some sides rely on positional dominance and sustained possession to create high-quality chances, while others favour vertical progression supported by strong defensive organisation. A third group prioritises intensity and wide play to generate shooting volume, whereas more reactive teams operate from deeper blocks and depend on transitions.

Overall, playing style does not directly determine league position: teams with contrasting approaches can achieve similar outcomes—such as Real Madrid and Atlético de Madrid (Figure A.8)—while stylistically similar teams may finish far apart, as seen with Real Betis and Getafe CF (Figure A.9)*.

*A more detailed version of this analysis is provided in the final section of Colab 1.

4. DATA PREPROCESSING

Before the clustering process, a variable selection step was conducted to reduce redundancy and mitigate potential multicollinearity effects. Correlation matrices (Figures A.10, A.11, A.12) were examined to identify highly correlated variables, as well as metrics representing the same tactical dimension — such as absolute counts and their corresponding ratios. In cases where multiple variables captured overlapping information, only one representation was retained to avoid duplication and preserve interpretability.

Particular emphasis was placed on retaining relative metrics, as these better reflect stylistic and efficiency-related aspects of performance rather than overall activity levels. By prioritizing proportional and rate-based measures, the analysis focuses on structural differences in playing style instead of distinctions driven primarily by raw volume. Following this selection process, the most representative variables from the attacking, build-up, and defensive phases were integrated into a single final dataset composed of 17 features.

Prior to model application, all variables were standardized to eliminate scale differences and ensure equal contribution to distance calculations. This step is especially important for distance-based clustering algorithms, as it prevents features with larger numerical ranges from disproportionately influencing the clustering solution and facilitates a more balanced identification of underlying performance patterns.

5. CLUSTERING MODELS APPLICATION AND RESULTS

This section presents the application of the selected clustering models to the final feature set and discusses the resulting groupings of teams.

5.1 K-Means Clustering

In Figure 4, the Elbow Method is applied to select the number of clusters for the K-Means algorithm.

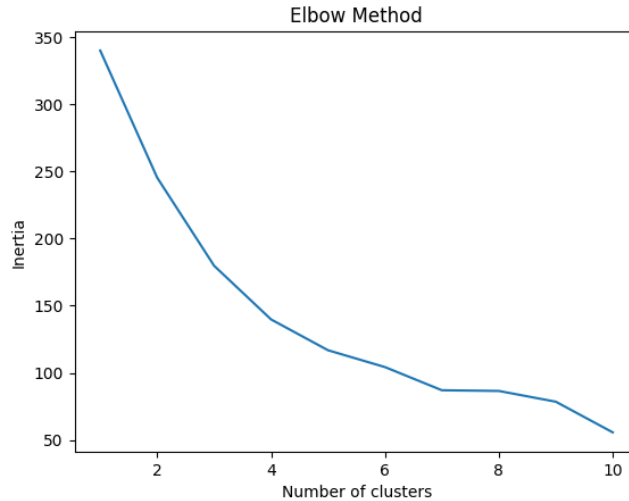


Figure 4. Elbow Method for Selecting the Number of Clusters

As shown in Figure 4, inertia decreases sharply up to $k = 4$, after which the curve begins to flatten. This indicates that adding more clusters does not lead to a significant improvement in clustering quality;

therefore, $k = 4$ is selected as the optimal number of clusters. To facilitate the interpretation of the resulting clustering structure, a PCA projection of the K-Means solution is presented in Figure 5. The first two principal components explain 47.7% and 13.1% of the total variance, respectively, accounting for 60.8% of the overall variability in the data.

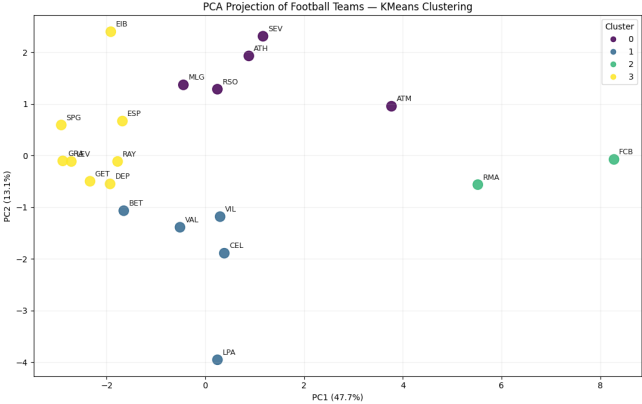


Figure 5. PCA Projection with K-Means Clustering

The visualization shows a reasonably clear separation between the clusters, primarily driven by the first principal component. FC Barcelona and Real Madrid stand out with high values on the first component, indicating performance profiles that are clearly differentiated from the rest of the teams. Although the remaining teams are more closely grouped, the separation between clusters remains well defined, with each cluster occupying a distinct region in the PCA space. This distribution suggests that the clustering captures meaningful differences among teams, even in cases where overall performance levels are more similar.

To interpret the clustering results obtained with the K-Means model, feature influence was assessed by analyzing the variability of cluster centroids across features.

At an individual feature level, Table 1 presents the top five variables with the highest centroid variability. The results indicate that shooting intensity, duel success, and passing-related variables dominate the separation between clusters. In particular, shots per 90 minutes, duel success rate, and long pass completion percentage emerge as the most discriminative features.

Variable	Centroid STD
Shots per 90 minutes	1.472
Duel won percentage	1.426
Long pass completion percentage	1.421
Total passes per 90 minutes	1.335
Goals per shot percentage	1.330

Table 1. Top 5 Influential Variables in K-Means

When feature importance is aggregated by tactical dimension, the results indicate a clear predominance of build-up variables, as reported in Table 2. Build-up metrics exhibit the highest average centroid

variability, followed by defensive variables, while attacking variables show comparatively lower aggregated variability. This pattern suggests that differences in possession structure and passing behavior contribute more strongly to the separation between clusters identified by the K-Means model, with defensive and attacking aspects providing additional but secondary differentiation.

Feature Group	Centroid STD
Build-up	1.165
Defense	0.993
Attack	0.978

Table 2. Feature Influence by Tactical Dimension in K-Means

5.2 Gaussian Mixture Models

Figure 6 shows the AIC and BIC values used to select the optimal number of components for the GMM. While the AIC reaches its minimum at $k = 5$, the BIC attains its lowest value at $k = 4$. In cases where the criteria disagree, the BIC is generally preferred due to its more conservative nature, as it favors simpler and more robust models. Therefore, a GMM with 4 clusters is chosen as the final configuration.

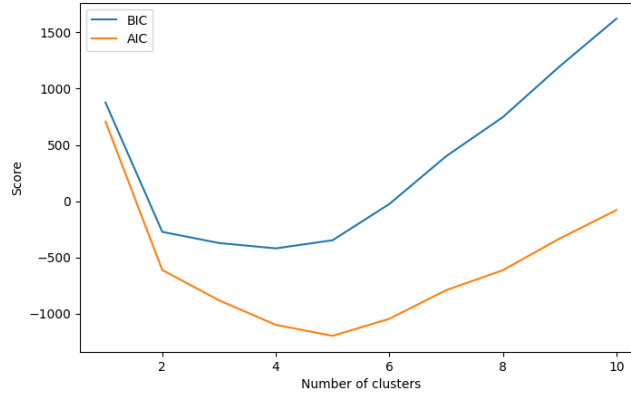


Figure 6. AIC and BIC for Selecting the Number of Clusters in GMM

To further examine the resulting cluster structure, Figure 7 presents the PCA projection of the teams under the GMM clustering solution, allowing for a visual assessment of the separation between clusters.

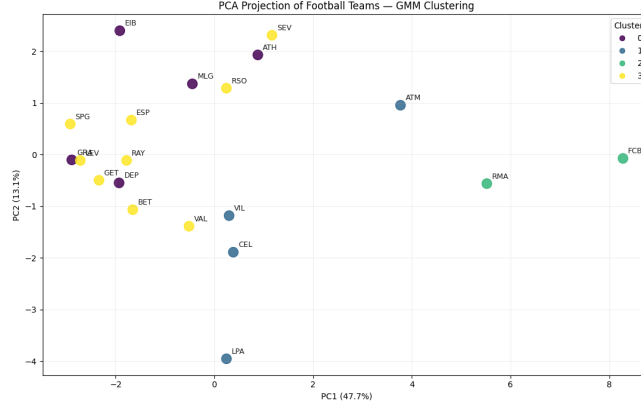


Figure 7. PCA Projection with GMM

As in the K-Means solution, FC Barcelona and Real Madrid remain clearly separated from the rest of the teams along the first principal component, indicating a consistent identification of these teams as a distinct performance group across clustering methods.

However, unlike K-Means, the remaining teams are distributed differently across clusters under the GMM framework. Although several clusters can still be distinguished, the separation between them is less pronounced. In particular, Clusters 0 and 3 exhibit noticeable overlap in the PCA space, with multiple teams occupying similar regions. This overlap suggests softer boundaries between groups, reflecting the probabilistic nature of GMM and indicating that some teams share mixed characteristics rather than belonging clearly to a single cluster.

To interpret the clustering results obtained with the GMM, feature influence was analyzed by examining the variability of the component means across clusters.

As shown in Table 3, the top five most influential variables are largely consistent with those identified in the K-Means model, with the same set of features appearing as the main drivers of cluster separation, although with minor changes in their relative ordering.

Variable	Mean STD
Shots per 90 minutes	1.445
Long pass completion percentage	1.428
Duel won percentage	1.414
Total passes per 90 minutes	1.363
Goals per shot percentage	1.332

Table 3. Top 5 Influential Variables in Gaussian Mixture Model

When feature importance is aggregated by tactical dimension, the results reported in Table 4 follow a pattern similar to that observed for K-Means, with build-up variables emerging as the most influential group. In contrast to the K-Means model, attacking variables display a slightly higher aggregated influence than defensive variables in the GMM, although the difference between both groups is small.

A comparable behavior was observed in K-Means, albeit with the relative ordering between attack and defense reversed.

Feature Group	Mean STD
Build-up	1.206
Attack	1.007
Defense	0.911

Table 4. Aggregated feature influence by tactical dimension in Gaussian Mixture Model clustering

5.3 Agglomerative Hierarchical Clustering

Figure 8 shows the dendrogram obtained using agglomerative hierarchical clustering with Ward linkage, where a clear increase in the fusion distances at higher levels suggests the presence of two well-defined clusters.

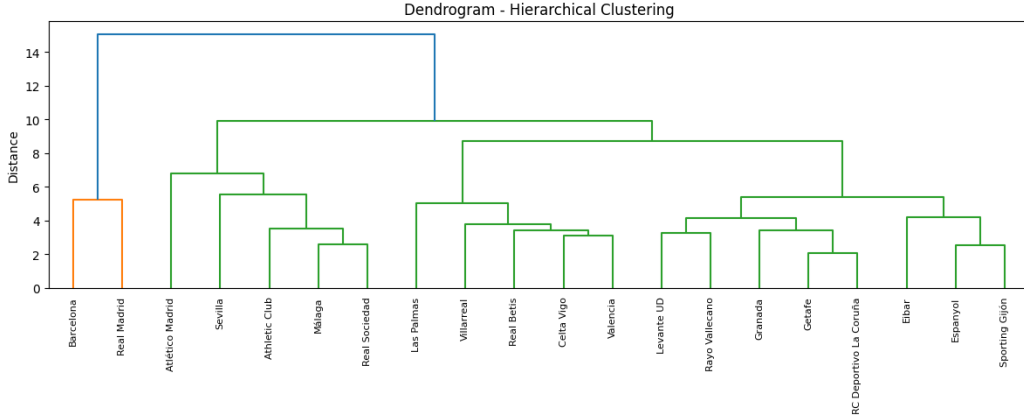


Figure 8. Dendrogram obtained using Agglomerative Hierarchical Clustering with Ward linkage

This observation is reinforced by the Silhouette Score shown in Figure 9. The score reaches its maximum value for $k = 2$ and decreases for larger numbers of clusters, indicating a loss of clustering quality. Taken together, both figures support the selection of two clusters as the optimal solution.

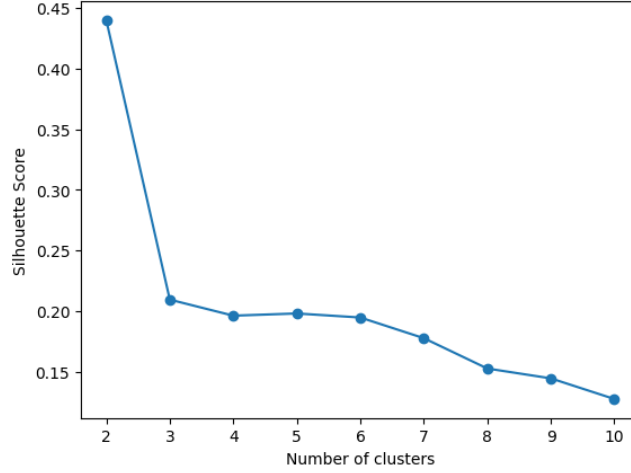


Figure 9. Silhouette Score for Selecting the Number of Clusters in Agglomerative Hierarchical Clustering

After selecting $k = 2$ as the optimal configuration for agglomerative hierarchical clustering, Figure 10 shows the corresponding PCA projection.

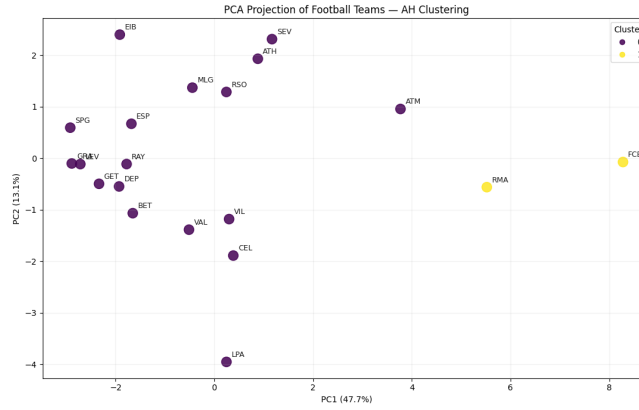


Figure 10. PCA Projection with Agglomerative Hierarchical Clustering

The results reveal a clear separation between FC Barcelona and Real Madrid and the rest of the teams, indicating that the hierarchical model primarily captures this dominant distinction. This configuration emphasizes the structural divide between the two top teams and the remainder of the league.

For the agglomerative hierarchical clustering, feature influence was assessed by comparing the mean values of each variable between the two resulting clusters.

At an individual feature level, the top five most influential variables (Table 5) are largely consistent with those identified in the K-Means and Gaussian Mixture Model analyses. The main difference lies in the presence of shots outside the box percentage instead of shots per 90 minutes, although both metrics capture shooting patterns.

When features are aggregated by tactical dimension (Table 6), build-up variables once again emerge as the dominant source of separation, with a substantially larger difference compared to attacking and

Variable	Mean Difference
Total passes per 90 minutes	182.123
Long pass completion percentage	17.951
Duel won percentage	11.719
Goals per shot percentage	6.399
Shots outside the box percentage	6.155

Table 5. Top 5 Influential Variables in Agglomerative Hierarchical Clustering

defensive variables, which display similar levels of influence. Overall, these results indicate that the main distinction between FC Barcelona and Real Madrid and the rest of the teams lies in differences in their build-up style of play.

Feature Group	Mean Difference
Build-up	41.081
Attack	4.656
Defense	4.374

Table 6. Aggregated feature influence by tactical dimension in Agglomerative Hierarchical Clustering

6. CONCLUSIONS

This study highlights the usefulness of unsupervised learning techniques for identifying performance patterns in football beyond traditional league standings. By applying K-Means, Gaussian Mixture Models, and Agglomerative Hierarchical Clustering to team-level data from the 2015/2016 La Liga season, the analysis reveals meaningful groupings of teams based on their tactical and performance characteristics.

A consistent result across all clustering methods is the clear separation of FC Barcelona and Real Madrid from the rest of the league. This distinction is primarily driven by build-up play variables, particularly passing volume and efficiency, which emerge as the most influential features in cluster formation. Defensive and attacking metrics play a secondary role, contributing to further differentiation but with lower overall impact.

K-Means and Gaussian Mixture Models both support a four-cluster structure that captures stylistic differences among teams, while the probabilistic nature of GMM reveals overlapping profiles for teams with mixed characteristics. In contrast, agglomerative hierarchical clustering emphasizes the most fundamental division in the data, isolating the two dominant teams and providing a highly interpretable global structure.

Overall, the findings indicate that, within this dataset and season, build-up play emerges as the key tactical dimension distinguishing team profiles in La Liga. This work demonstrates the potential of clustering techniques as effective exploratory tools in football analytics, with applications in tactical analysis, team benchmarking, and performance profiling. Future research could extend this approach

by incorporating additional seasons or more granular data to further validate and refine the identified patterns.

APPENDIX A. FIGURES

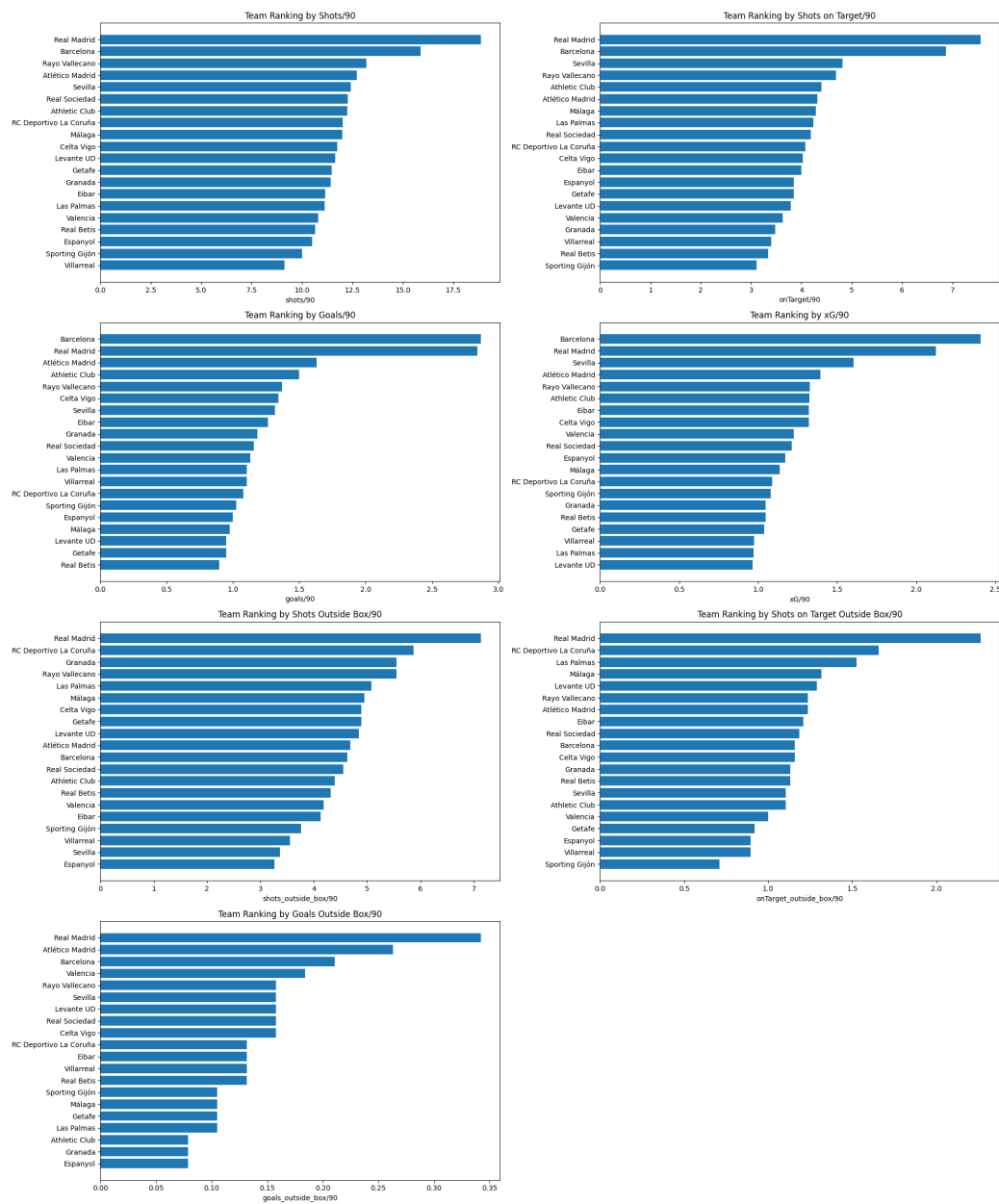


Figure A.1. Histograms of Attacking Variables

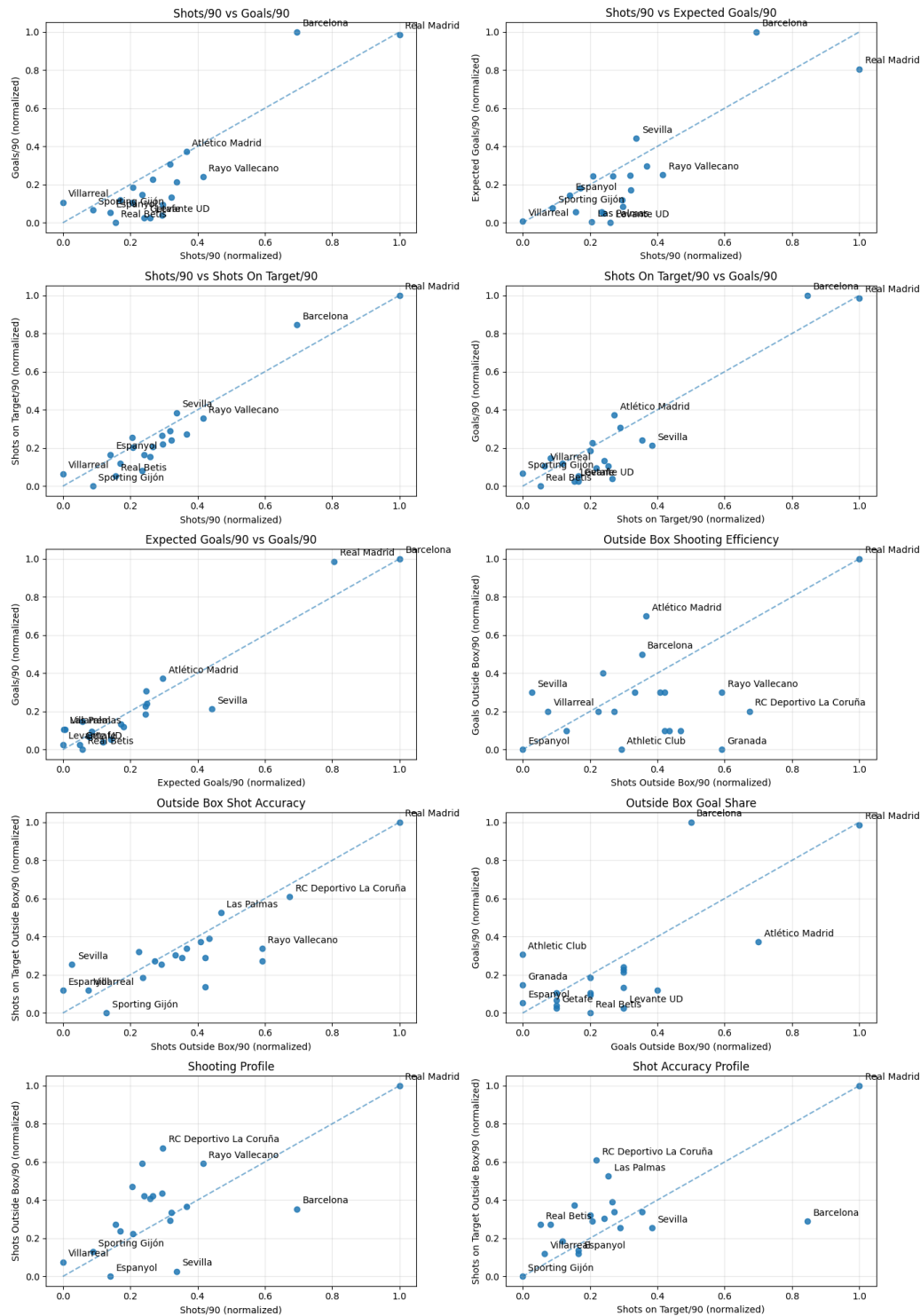


Figure A.2. Scatterplots of Attacking Variables

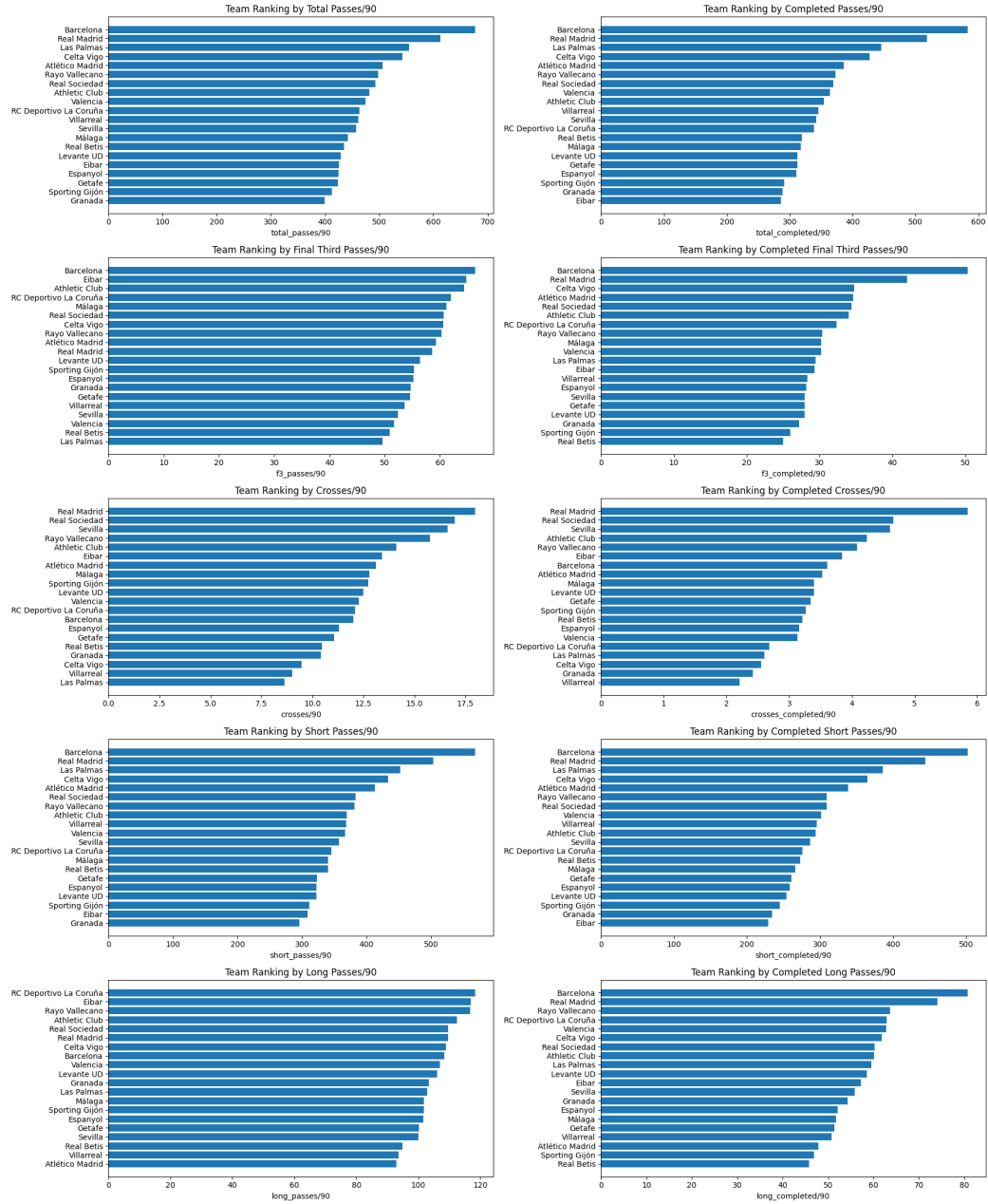


Figure A.3. Histograms of Build-Up Variables

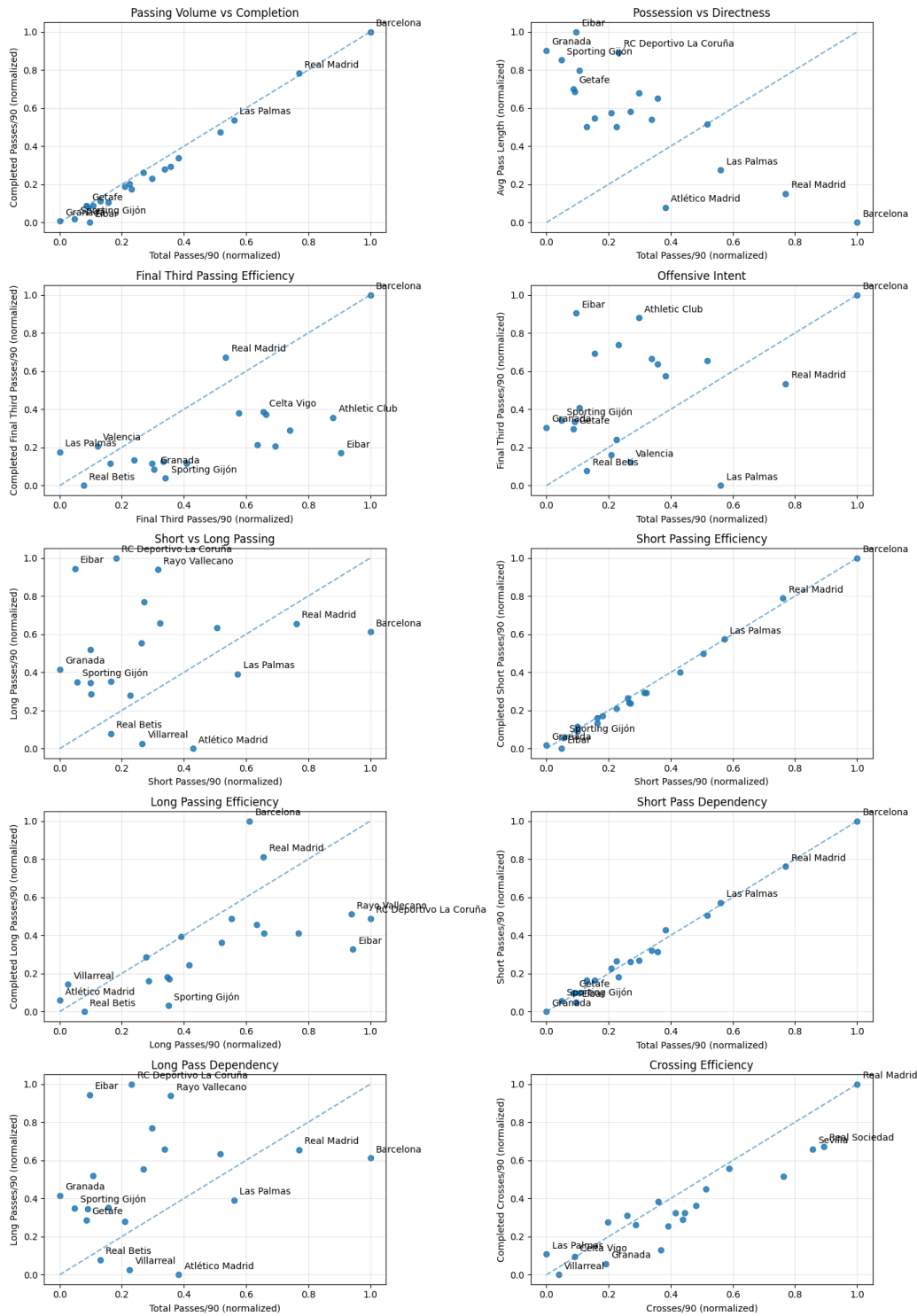


Figure A.4. Scatterplots of Build-Up Variables



Figure A.5. Histograms of Defensive Variables

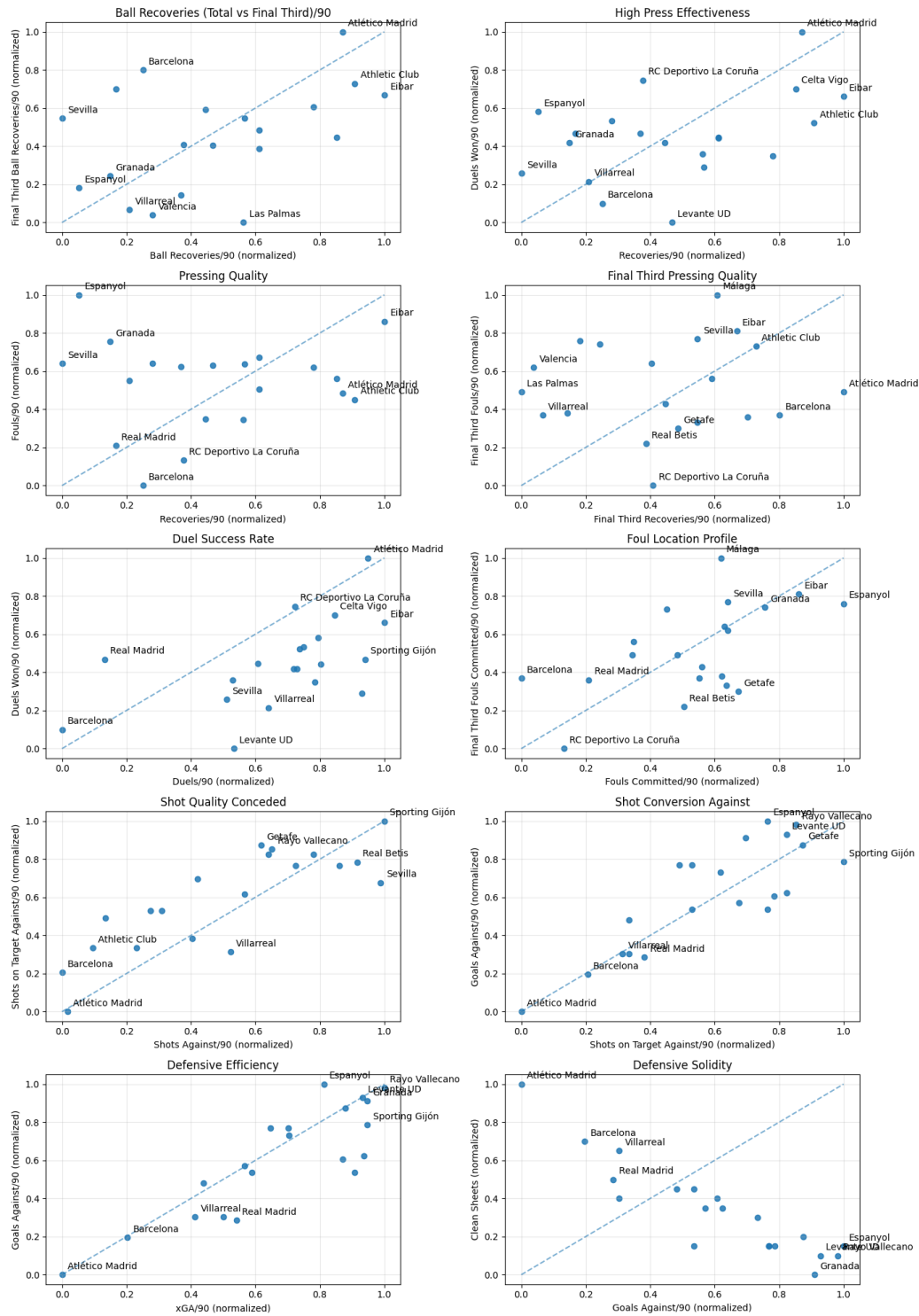


Figure A.6. Scatterplots of Defensive Variables

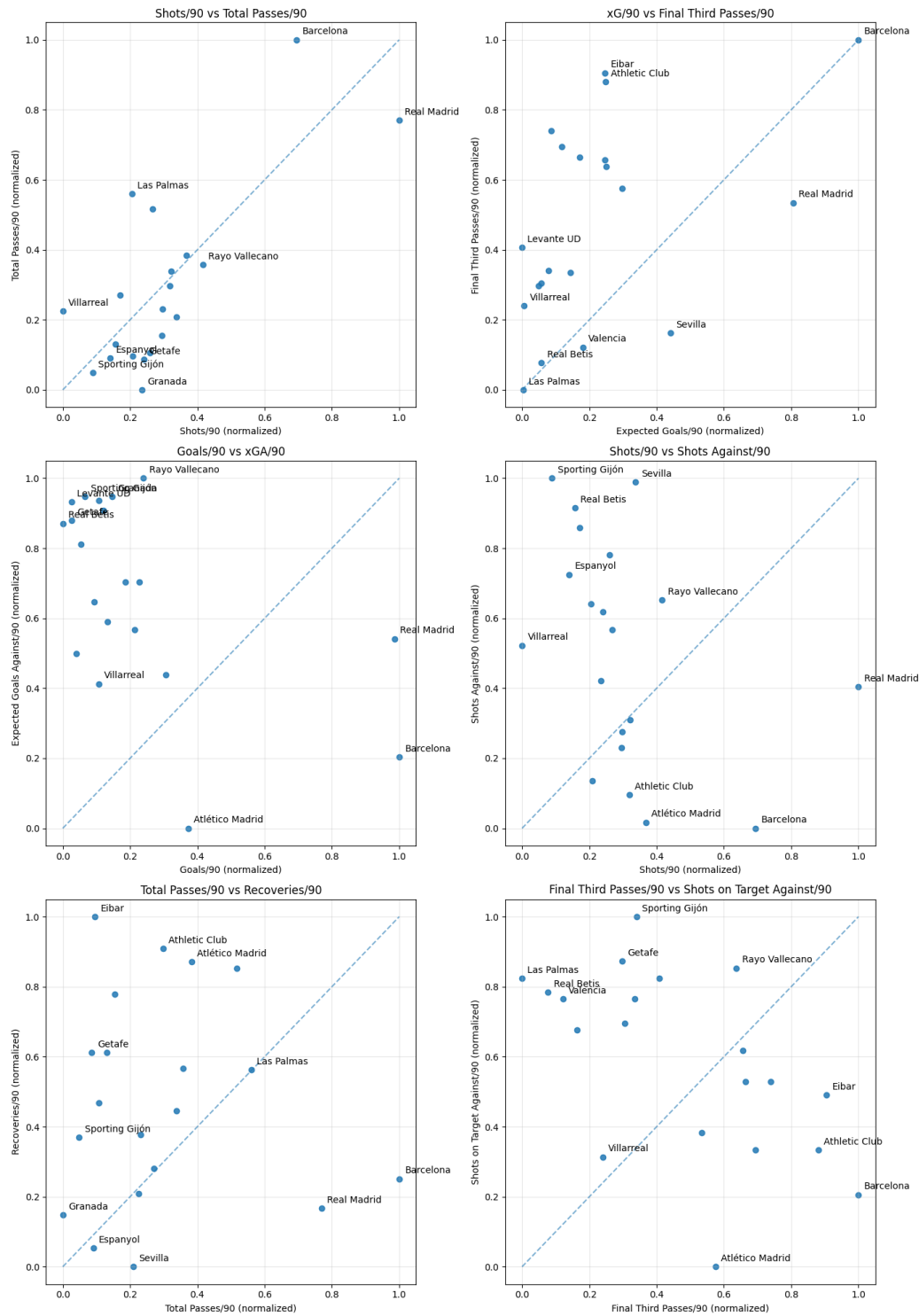


Figure A.7. Scatterplots of Multi-Phase Performance Variables

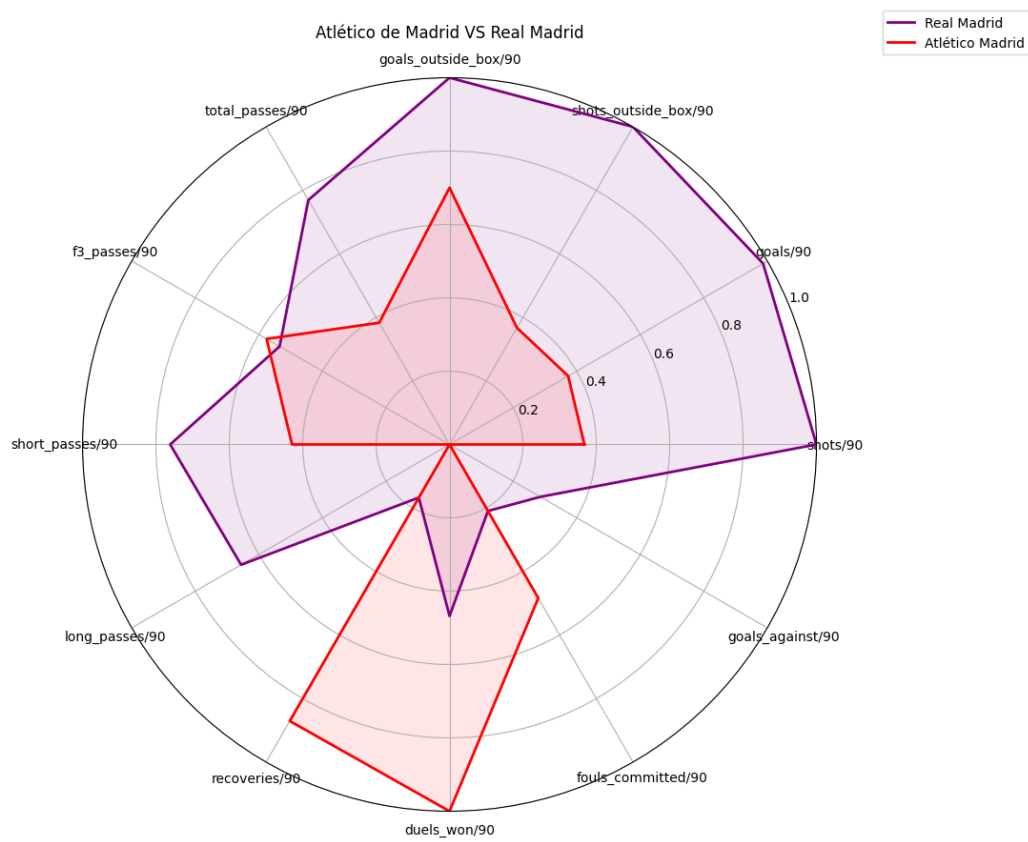


Figure A.8. Radar Plot Atlético de Madrid vs Real Madrid

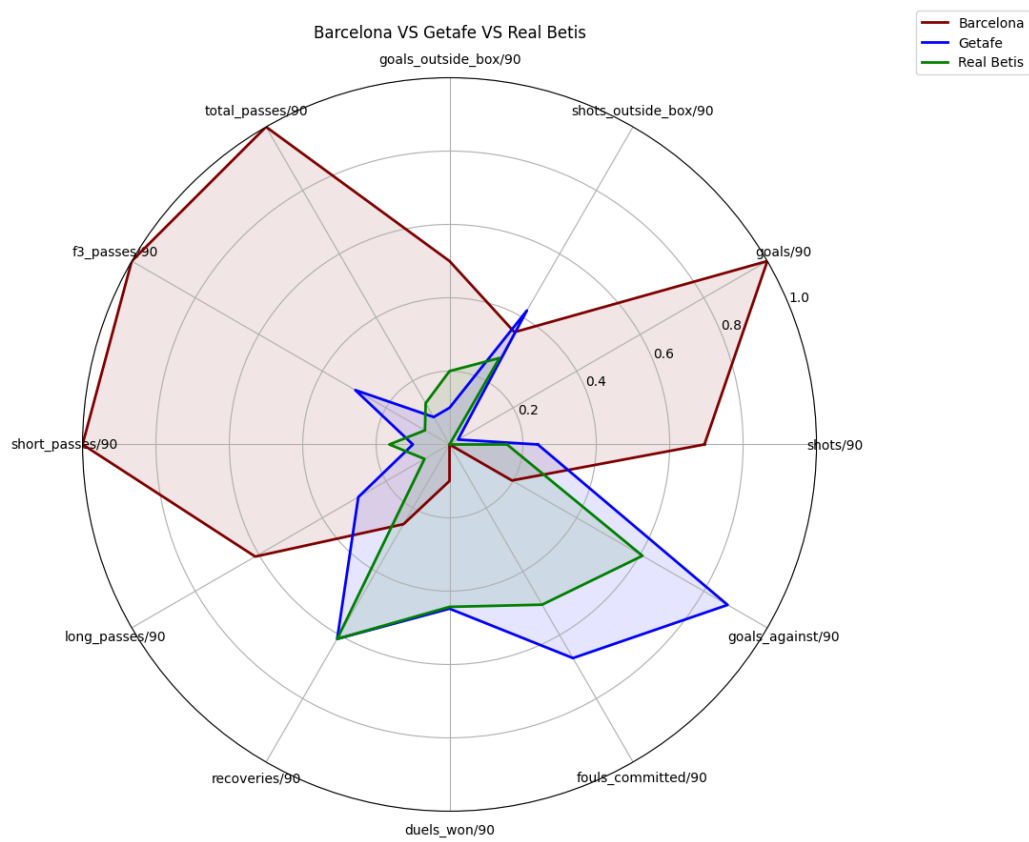


Figure A.9. Radar Plot Barcelona vs Real Betis vs Getafe

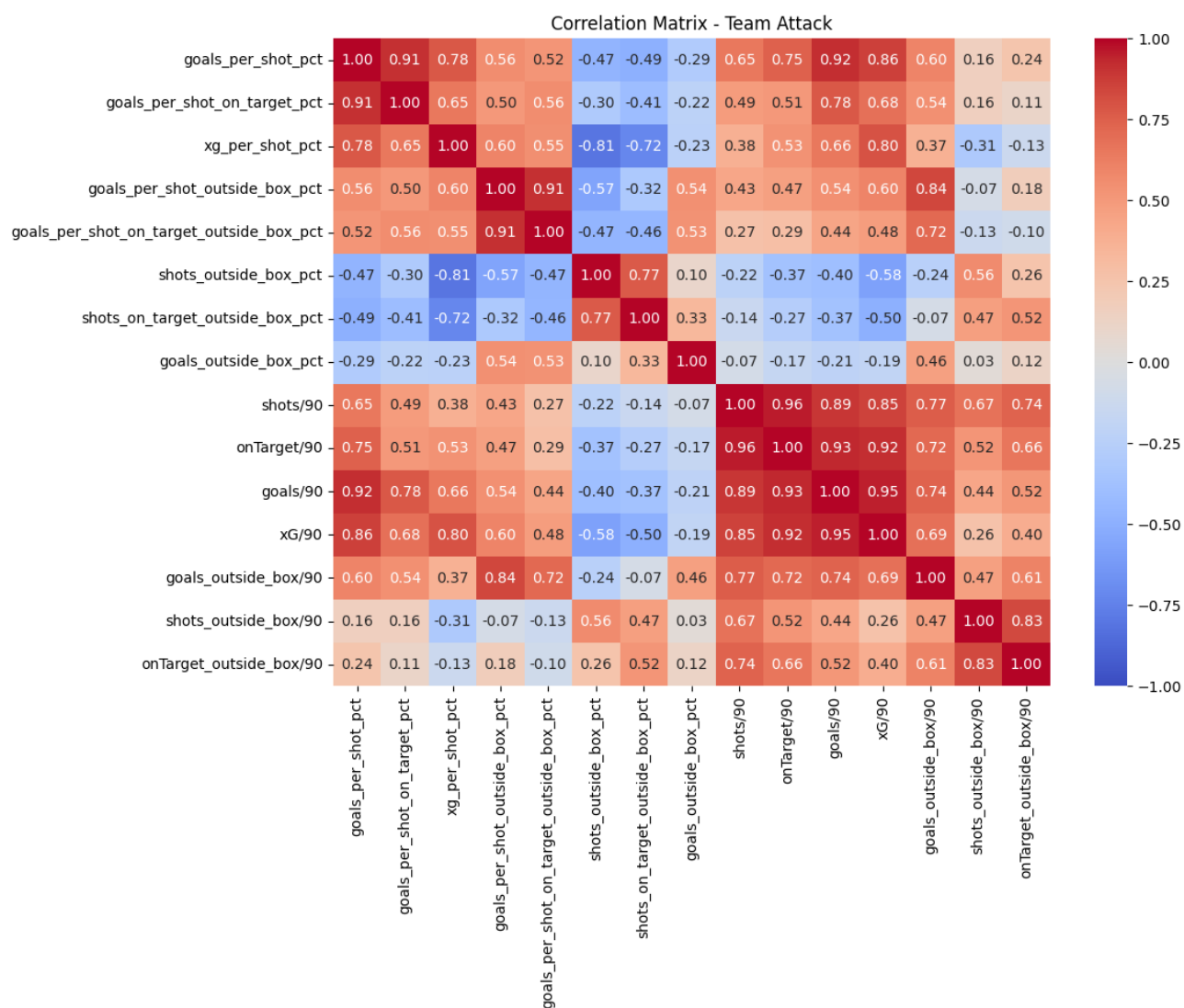


Figure A.10. Correlation Matrix of Attacking Variables

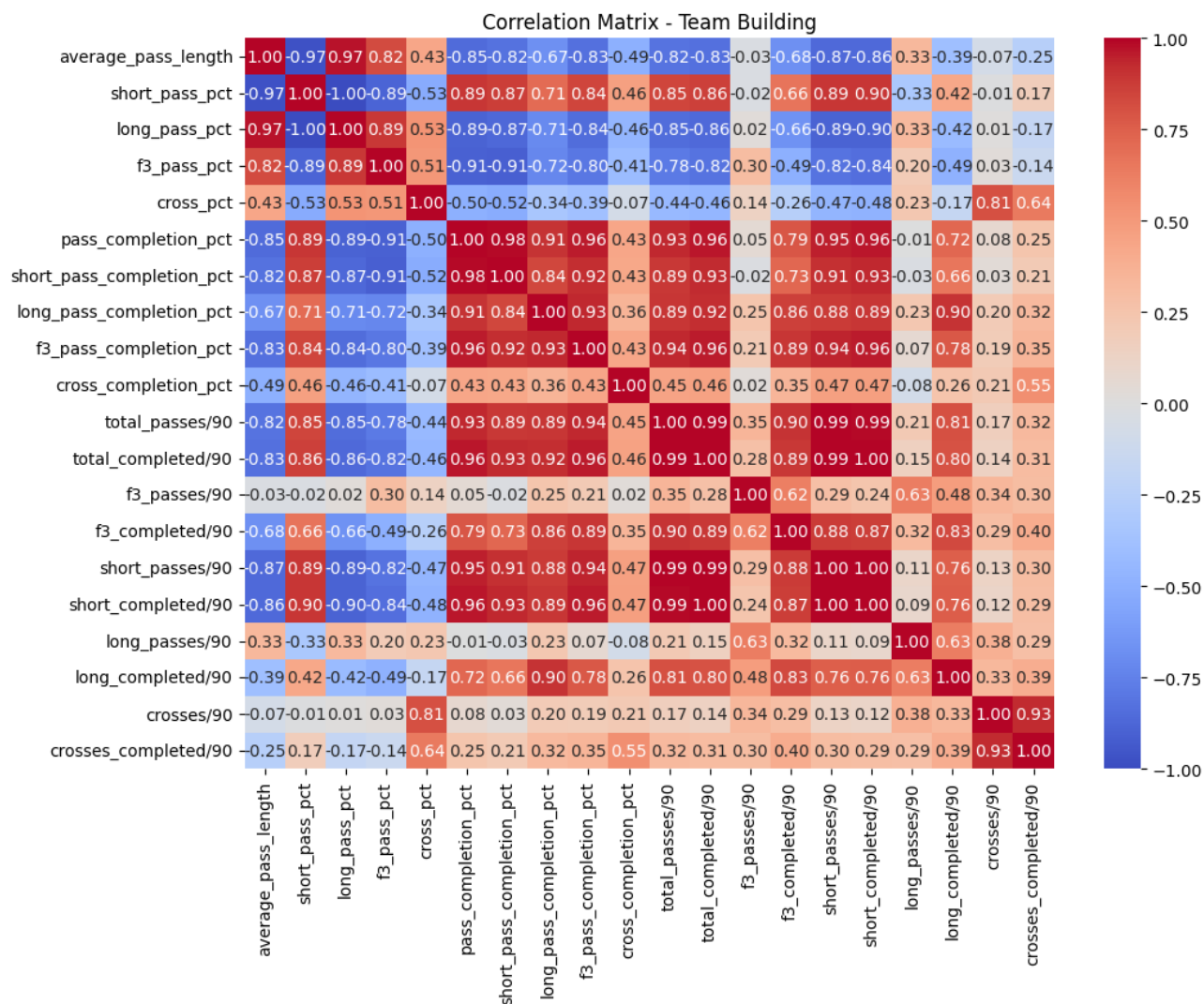


Figure A.11. Correlation Matrix of Build-Up Variables

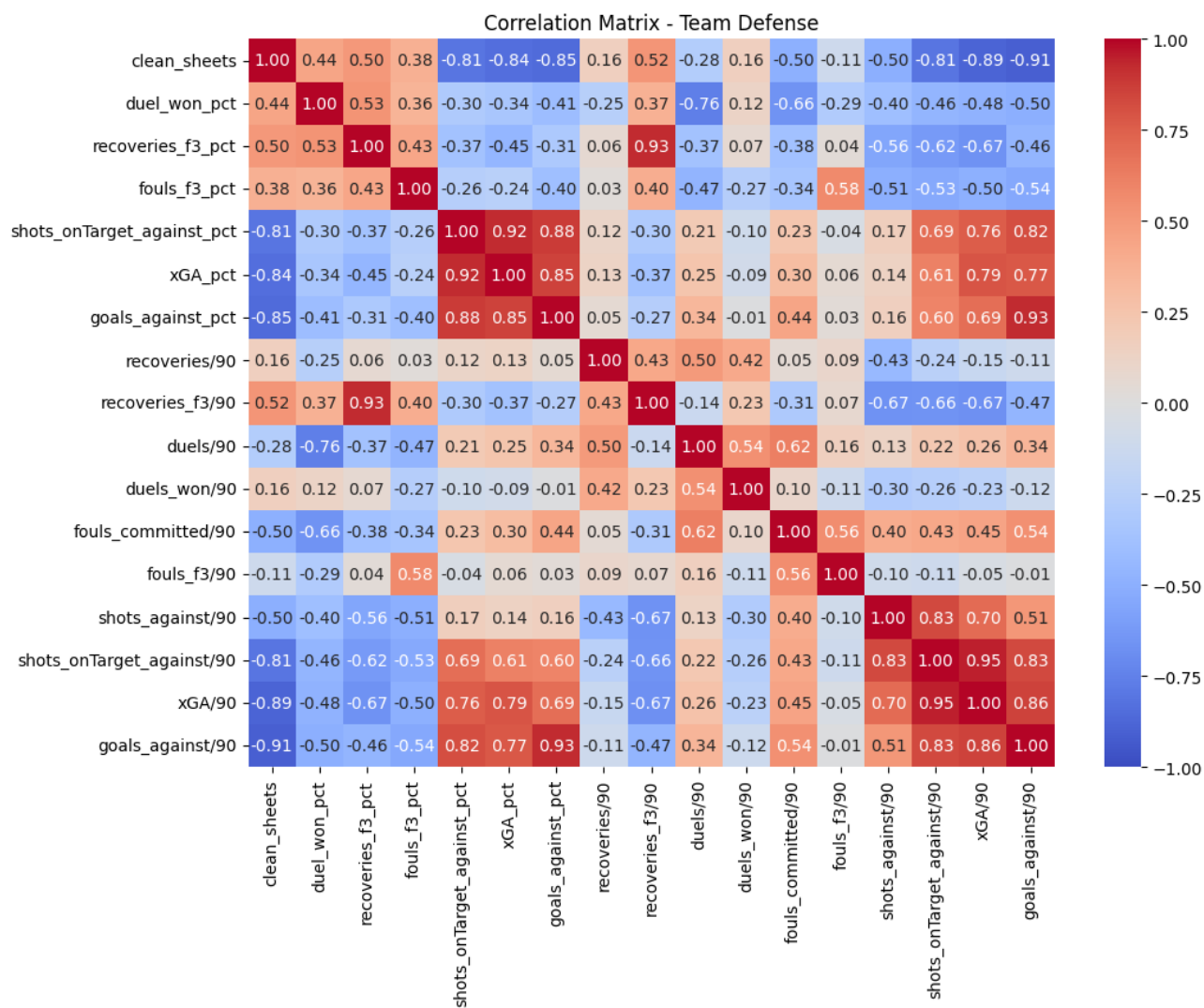


Figure A.12. Correlation Matrix of Defensive Variables

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